

FROM MODEL DIFFS TO CAUSAL CONTROL

A shared sparse representation for explaining and testing behavioral differences between code models

BEHAVIOR



FEATURES



STEERING

RESEARCH QUESTION

Evaluation tells us what changed. CrossCoder asks which internal differences can explain—and control—it.

BEHAVIOR

Which tasks changed?

pass → fail

fail → pass

What failure mode changed?



REPRESENTATION

Paired layer-16 residuals

Same-text alignment

Shared sparse features



CAUSAL TEST

Choose candidate directions

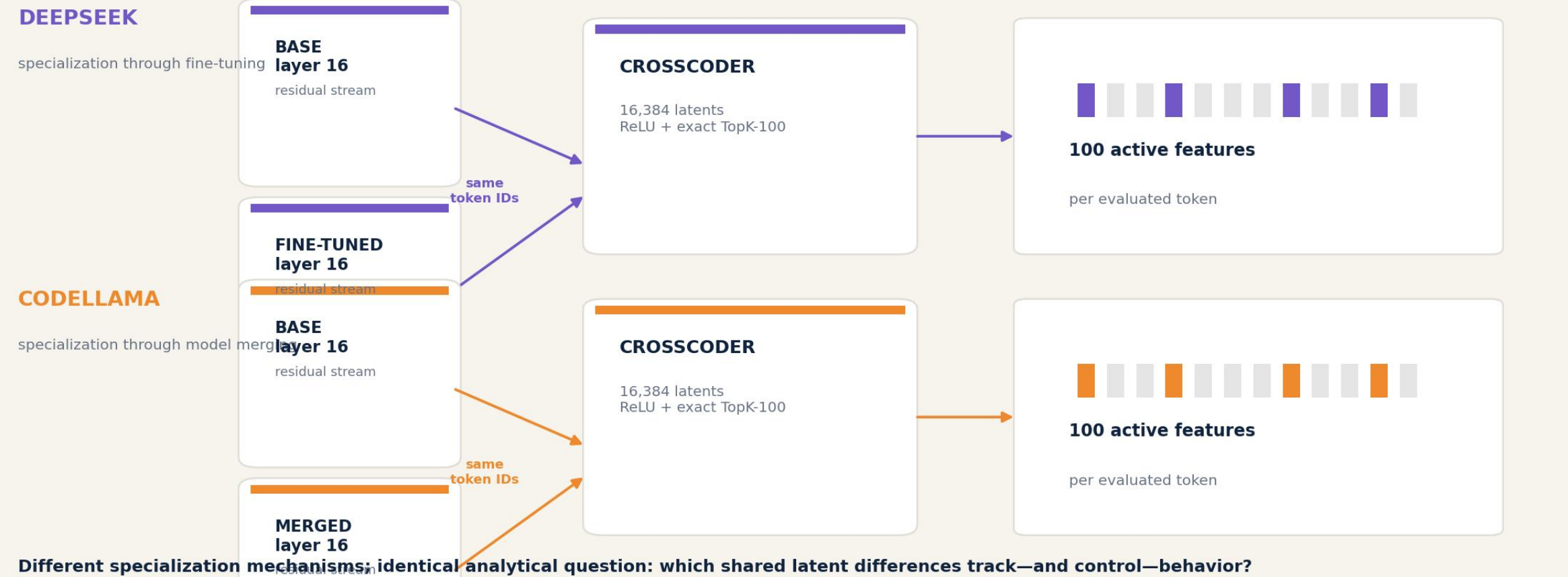
Intervene during generation

Compare against controls

Behavior localizes the model difference. Sparse features turn it into a testable hypothesis.

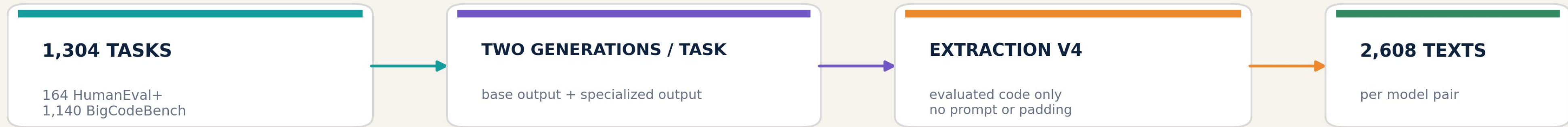
ONE ANALYSIS FRAMEWORK, TWO VERY DIFFERENT MODEL CHANGES

Same token IDs turn each model pair into a shared sparse vocabulary

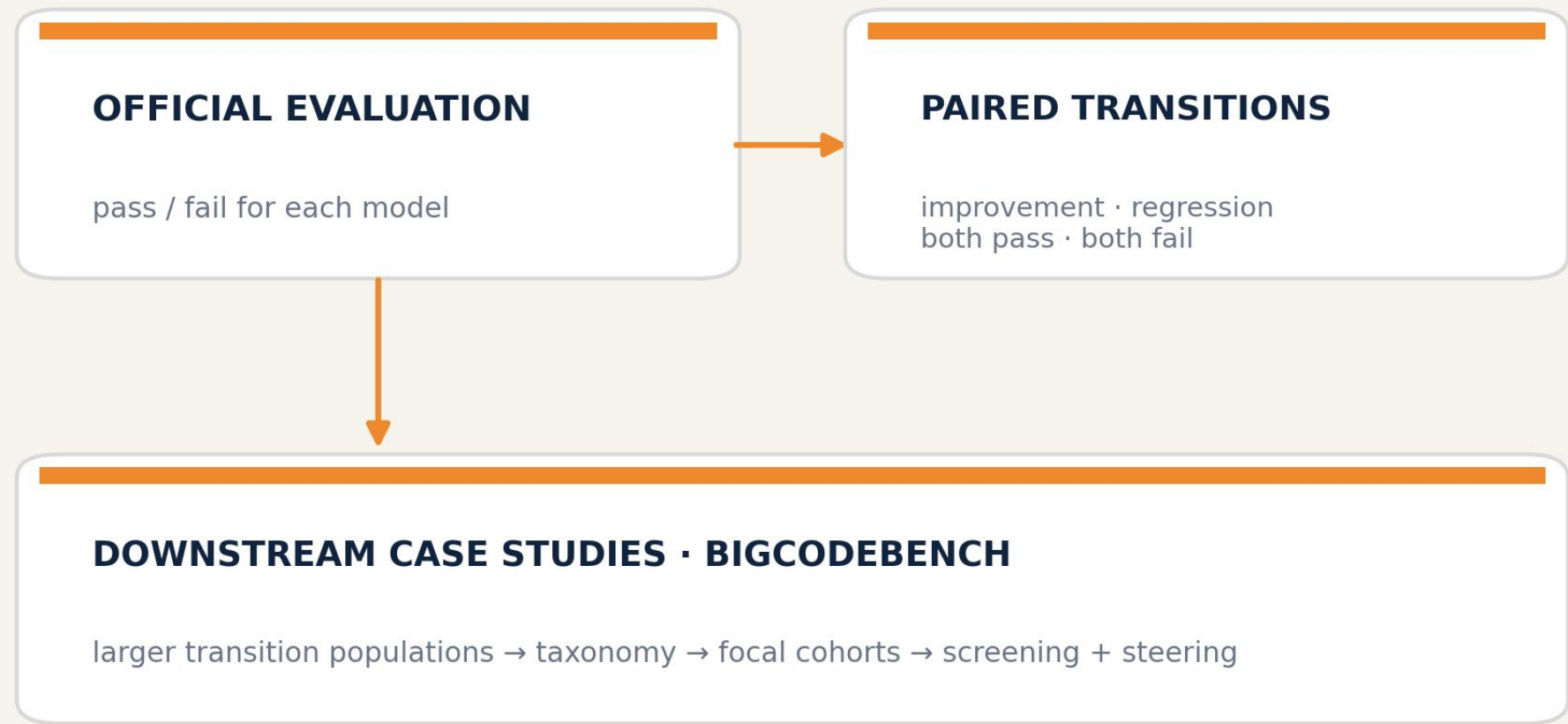


FROM BENCHMARK GENERATIONS TO A SHARED SPARSE REPRESENTATION

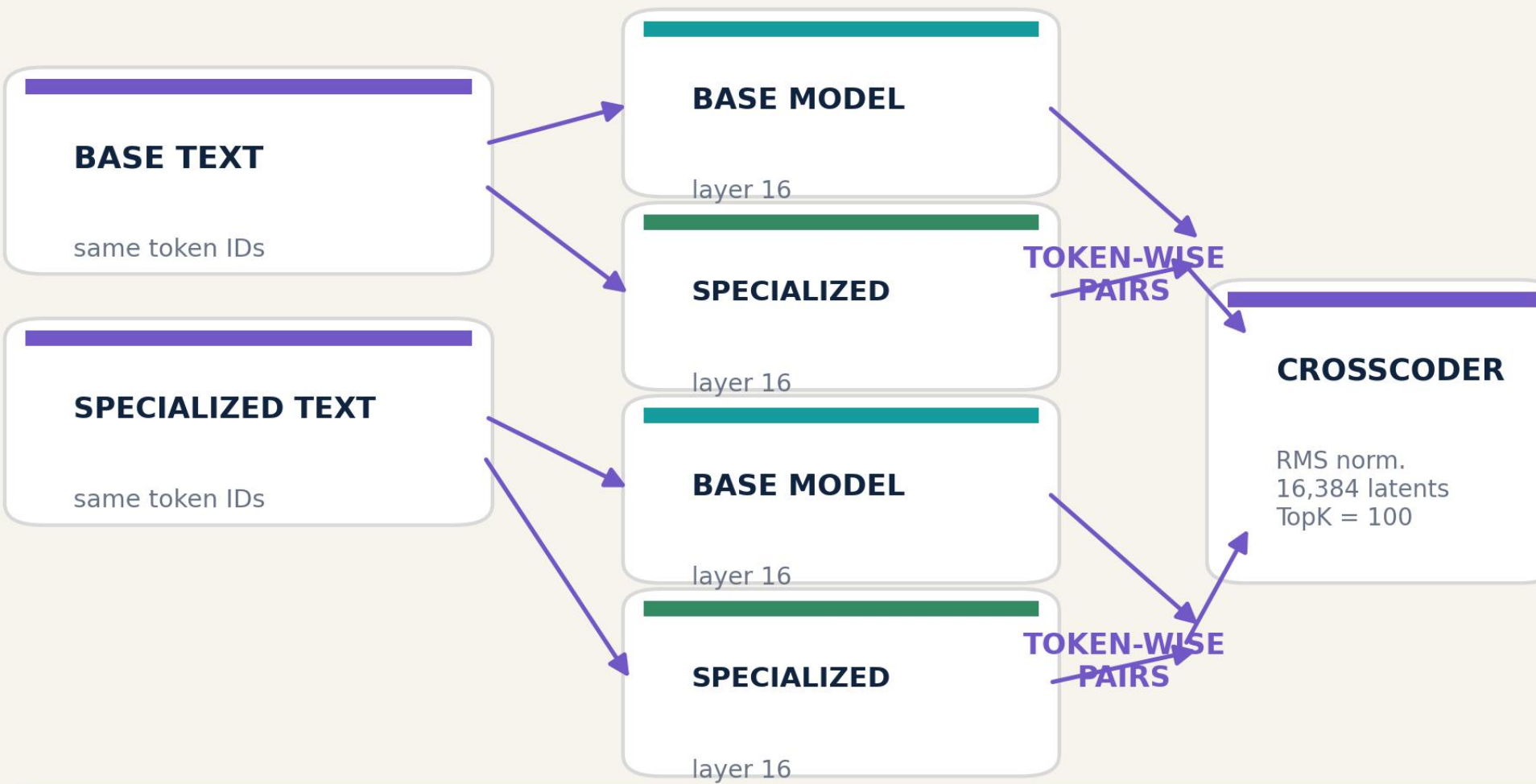
Evaluation and same-text replay reuse the same extracted code for different purposes



A BEHAVIORAL LABELS



B SAME-TEXT ACTIVATIONS



Every activation pair represents the same token at the same textual position.

THE TWO SPECIALIZATIONS MOVE PERFORMANCE IN OPPOSITE DIRECTIONS

Canonical extraction-v4 labels · same raw generations, audited reprocessing

DEEPSEEK · BIGCODEBENCH

268/1140

BASE

404/1140

FINE-TUNED

+11.93 pp

BigCodeBench
pass-rate change

CODELLAMA · BIGCODEBENCH

314/1140

BASE

27/1140

MERGED

-25.18 pp

BigCodeBench
pass-rate change

DeepSeek: 215 improvements · 79 regressions

CodeLlama: 4 improvements · 291 regressions

RECATCHER CONTEXT

Directionally consistent motivation: specialization can improve aggregate capability while still creating localized regressions.

Not a direct numerical replication: ReCatcher used 10 generations/prompt and different regression definitions.

WE FOCUS ON TWO ONE-SIDED BEHAVIORAL TRANSITIONS

Taxonomy turns aggregate model differences into semantically targeted cohorts

DEEPSEEK · 215 IMPROVEMENTS

base fail → fine-tuned pass

119/215: generated test/import contamination

48: wrong output or logic

16: missing name/import

32: other primary categories

Mechanistic focus: 80 contamination cases

CODELLAMA · 291 REGRESSIONS

base pass → merged fail

120/291: API/type mismatch

50: wrong logic/other runtime

43: edge case/exception

78: generation, import, syntax, commentary

Mechanistic focus: 50 wrong-logic/runtime cases

Improvements: only 4 positives · EXCLUDED

HOW WE CLASSIFIED

Rule/AI-assisted taxonomy + human spot checks

IMPORTANT QUALIFICATION

Primary = most salient observed error—not a unique root cause. CodeLlama failures are heterogeneous and often

DEEPSEEK /5 · CONTAMINATION

BASE · FAIL

Correct-looking body continues into:

```
`#tests/test_task_func.py`  
`import pytest`
```

FINE-TUNED · PASS

Completes the function and stops after a short example.

First textual divergence $\approx 76\%$ of shorter code;
target boundary is late/post-solution.

CODELLAMA /490 · WRONG LOGIC/API CONTRACT

BASE · PASS

parses XML $\rightarrow$ writes JSON file $\rightarrow$ returns result

MERGED · FAIL

```
`return xmltodict.parse(s)`  
omits the required file write side effect
```

First divergence $\approx 81\%$ of base code;
target region is the late function body.

FEATURE SCREENING

We rank stable behavioral contrasts—not raw activation alone



Minimum support

max(3 tasks, 10% of positives)

Interpretation

E/V is permutation signal-to-noise—not a z-score or p-value.

Positive and negative orientations are retained: either enrichment direction may support behavior-aligned steering.

SCREEN FIRST; TEST CAUSALLY AT A COMMON OPERATING POINT

The statistical shortlist reduces the intervention search by ~99.8%

16,384

latents / CrossCoder

≈ 30-40

candidates / model

≈ 0.2%

of latent space

WHY $|\alpha| = 3$?

Earlier dose-response work identified a practical exploratory point: visible causal changes below the most disruptive magnitude.

WHY STANDARDIZE?

Optimizing on a single metric makes dozens of candidates comparable before selecting five features for com

$|\alpha|=3$ is a common operating point—not an independently validated optimum.

WHAT THE SCREENING ACTUALLY RETURNS

Representative valid cells; E/V is the primary ranking signal

DEEPSEEK · IMPROVEMENT ASSOCIATION

Rank	Feature	Effect	E/V
1	3602	-0.021	-8.23
2	10168	-0.018	-7.67
3	5422	-0.005	-7.66
4	1982	-0.015	-7.53
5	1862	-0.008	-7.45
6	1650	-0.026	-7.28
7	5039	-0.088	-7.26
8	9580	-0.013	-7.24
9	5837	-0.010	-7.18
10	9537	-0.038	-7.15

CODELLAMA · REGRESSION ASSOCIATION

Rank	Feature	Effect	E/V
1	13147	+0.126	+5.24
2	8992	-0.011	-4.93
3	1840	-0.046	-4.77
4	16161	+0.127	+4.76
5	15471	+0.005	+4.75
6	10570	+0.188	+4.60
7	8138	-4.973	-4.51
8	7353	+0.228	+4.48
9	14359	-0.166	-4.46
10	2825	+0.338	+4.38

Examples—not a single global ranking. Candidate pools are formed by union and deduplication across valid cells.

ERROR-FOCUSED TOP-10 LISTS DEFINE THE NEW $\alpha=3$ CANDIDATE POOL

Three summaries per cell: max · mean · active fraction; ranking by absolute E/V

DEEPSEEK · 80 CONTAMINATION IMPROVEMENTS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
13879	14175	2913	2468
6895	9666	7728	14295
1078	1206	15821	8661
596	8686	12308	7728
8185	7169	16264	2913
14820	11612	2957	2957
1939	7804	3602	9700
15235	1939	2621	6305
15733	4994	6877	10736
16205	1591	14006	3602

40 nominations → 35 unique features

5 feature(s) appear in more than one list

CODELLAMA · 50 WRONG-LOGIC REGRESSIONS

ASSOC · GLOBAL	ASSOC · LOCAL	PAIRED · GLOBAL	PAIRED · LOCAL
7692	15559	8875	4173
10818	3097	7588	15559
5642	565	12895	12623
447	4173	447	15968
11596	15890	4634	8142
1017	414	5862	9419
4634	4740	3366	13231
7588	13768	7486	414
3442	4309	1017	13151
8384	14969	5642	13172

40 nominations → 32 unique features

8 feature(s) appear in more than one list

NEW $\alpha=3$ SWEEP: exactly 35 DeepSeek + 32 CodeLlama unique features · no post-hoc additions

Hard gate: baseline must reproduce the original raw completion byte for byte before any steered arm runs.

SCREEN FIRST; TEST CAUSALLY AT A COMMON OPERATING POINT

The statistical shortlist reduces the intervention search by ~99.8%

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≈30-40

candidates / model



≈0.2%

of latent space

WHY $|\alpha| = 3$?

Earlier dose-response work suggested a practical exploratory point: visible causal changes below the most disruptive high-dose regime

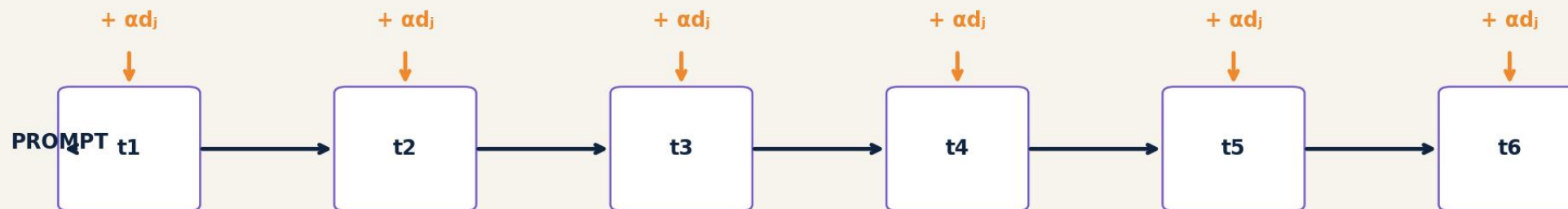
WHY STANDARDIZE?

On a logarithmic scale, the range of candidates comparable before full dose curves, reverse-model t

$|\alpha|=3$ is an operating point—not an independently validated optimum.

OUR CURRENT INTERVENTION IS CONTINUOUS STEERING

The selected decoder direction is added at every autoregressive generation step



CURRENT PROTOCOL

Layer 16 · last residual position
feature decoder on intervened model side
applied through the full generation

WHAT IT IS NOT

Not a one-token intervention
not activation-gated or clamped
not conditional on natural activation

This tests continuous control of the generation trajectory—not whether one naturally active token alone caused the error.

THE COMPLETE RESULT SEPARATES A STRONGER AND A WEAKER REGIM

Official BigCodeBench 0.1.5 evaluation · $\alpha=3$ · continuous last-token steering

DEEPSEEK

Features evaluated 35/35
Baseline 0/80
Feature-task transitions 116
Unique corrected tasks 16/80

Several features produce repeated official corrections.

CODELLAMA

Features evaluated 32/32
Baseline 0/50
Feature-task transitions 6
Unique corrected tasks 4/50

Single-feature corrections remain sparse.

STRONGER / LOCALIZED

DeepSeek contains several multi-task causal candidates.

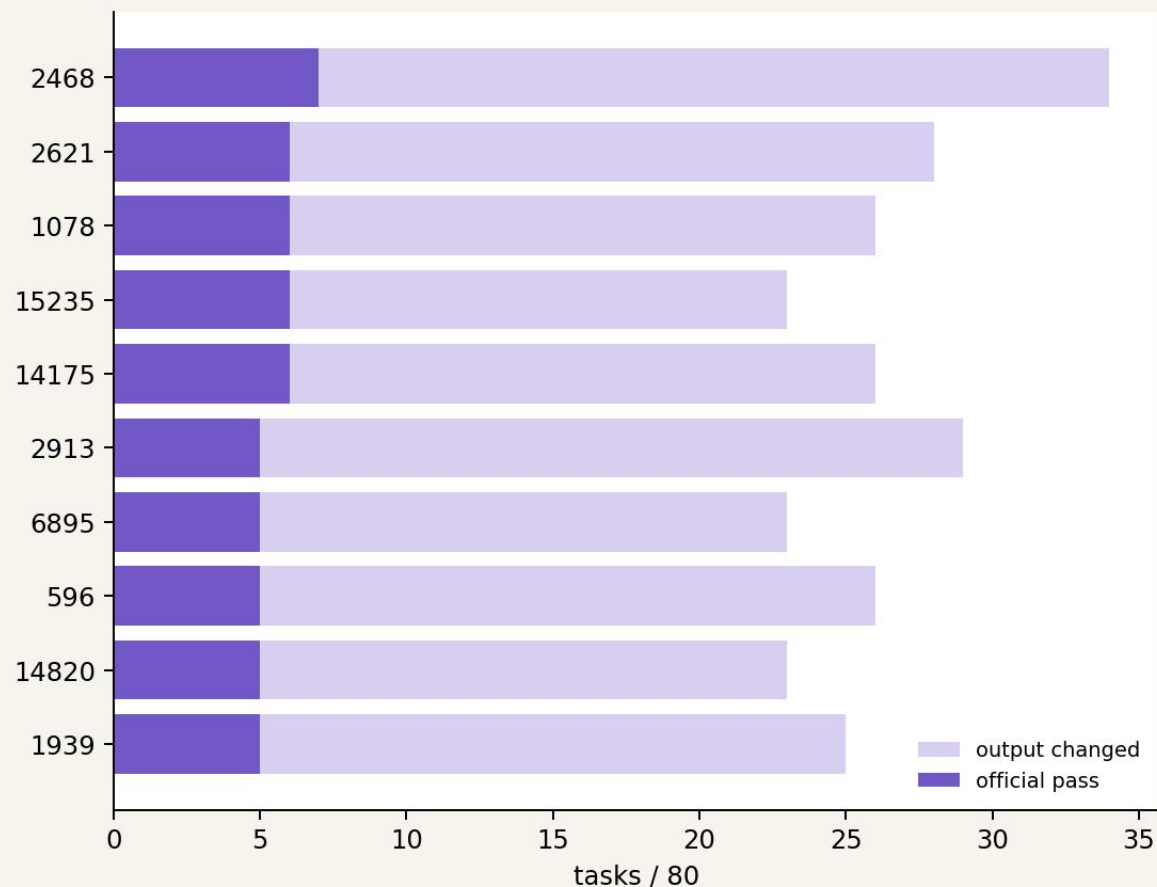
WEAKER / DIFFUSE

CodeLlama yields sparse, task-specific corrections.

COMPLETE · DeepSeek 35/35 and CodeLlama 32/32; canonical $\alpha=0$ gates passed before steering.

DEEPSEEK: THE COMPLETE $\alpha=3$ RANKING ADDS TWO TOP-5 FEATURES

Causal outcome first; |E/V| breaks ties among features with equal official passes



FINAL TOP 5

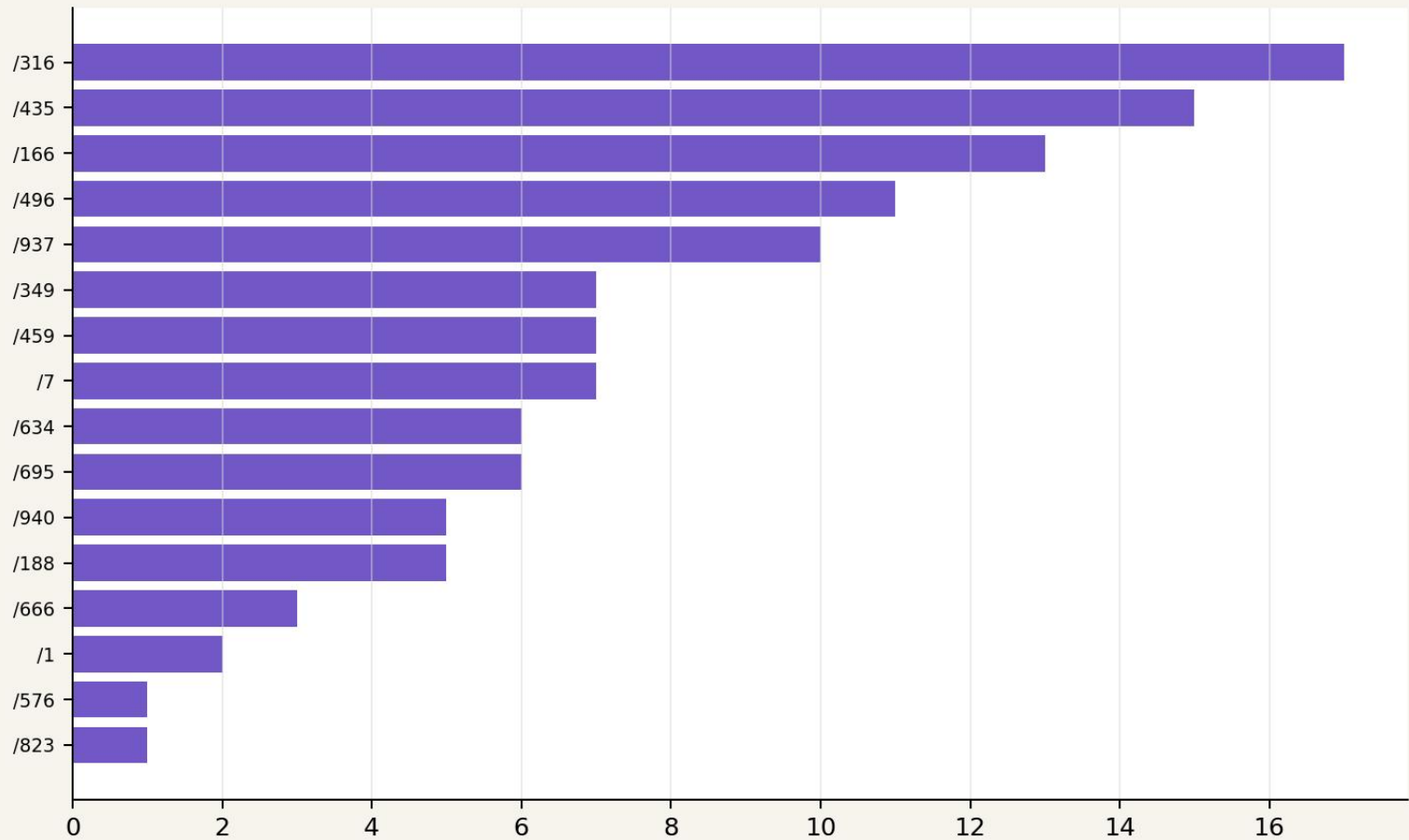
1. 2468 7/80 pass 34/80 changed E/V -7.29
2. 2621 6/80 pass 28/80 changed E/V -10.53
3. 1078 6/80 pass 26/80 changed E/V +5.72
4. 15235 6/80 pass 23/80 changed E/V -4.61
5. 14175 6/80 pass 26/80 changed E/V +4.09

1078 and 14175 enter after completion of all 35 arms.

2913 is #6 at $\alpha=3$, but later peaks at 11/80 at $\alpha=-5$.

DEEPSEEK: 116 TRANSITIONS COLLAPSE TO 16 SUSCEPTIBLE TASKS

Feature-task outcomes reveal both latent leverage and task-level susceptibility



16 unique corrections across 80 contamination-focused improvements.

CONCENTRATION

- /316 17 features
- /435 15 features
- /166 13 features
- /496 11 features
- /937 10 features

NEXT COMPARISON

The complete dose experiment tests whether the selected pool reaches co

FEATURE 2468 IS THE LEADING DEEPSEEK CANDIDATE

Meaning ↔ failure mode ↔ activation timing must remain distinct

SCREENING

paired local · rank 1
E/V -7.29
summary: max
selected around the late divergence window

CAUSAL OUTCOME

7/80 official passes
34/80 outputs changed
13/80 contamination cleanups
baseline: 0/80

CAUSED TIMING

Among changed outputs:
mean first divergence: 42.7%
median: 40.8%

Continuous steering can redirect earlier than the natural activation window

SUPPORTED local screening nominated a feature with repeated official causal effects.

NOT YET SUPPORTED feature-specificity versus sham/matched latent controls.

CODELLAMA: SINGLE-FEATURE CONTROL IS WEAK AND TASK-SPECIFIC

Complete 32-feature sweep; baseline 0/50

SIX FEATURES · ONE PASS EACH

7692 → BigCodeBench/119
10818 → BigCodeBench/119
5642 → BigCodeBench/630
11596 → BigCodeBench/119
4309 → BigCodeBench/490
8142 → BigCodeBench/823

FOUR UNIQUE TASKS

/119 corrected by 3 features
/823 corrected by 1
/490 corrected by 1
/630 corrected by 1

26/32 features produced no official correction.

Trajectory changes occur earlier and more diffusely than in the contamination-focused DeepSeek cohort.

Boundary condition: correlation and trajectory control do not imply that one latent can repair a heterogeneous merged-model regression.

NEXT TARGETS COMBINE CAUSAL EFFECT WITH SCREENING STRENGTH

Ordering: official passes first; |E/V| breaks ties

DEEPSEEK

RANK	FEATURE	PASS	BEST SCREEN
1	2468	7/80	paired local · #1
2	2621	6/80	paired global · #8
3	15235	6/80	association global · #8
4	14175	6/80	association local · #1
5	2913	5/80	paired global · #1

CODELLAMA

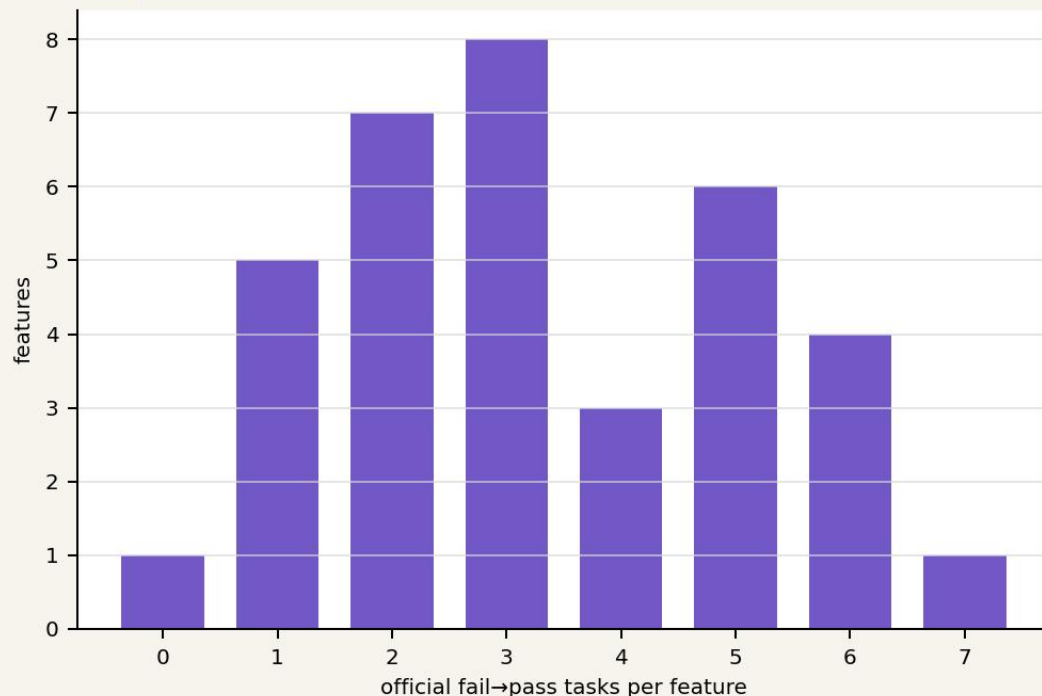
E/V	RANK	FEATURE	PASS	BEST SCREEN	E/V
7.29	1	7692	1/50	association global · #1	5.90
10.53	2	10818	1/50	association global · #2	5.90
4.61	3	5642	1/50	association global · #3	5.74
4.09	4	11596	1/50	association global · #5	5.61
10.95	5	4309	1/50	association local · #9	4.86

These are candidates for full dose curves plus sham and randomly selected/matched latent controls—not confirmatory winners yet.

THE $\alpha=3$ SWEEP REVEALS BOTH MODEL AND TASK SENSITIVITY

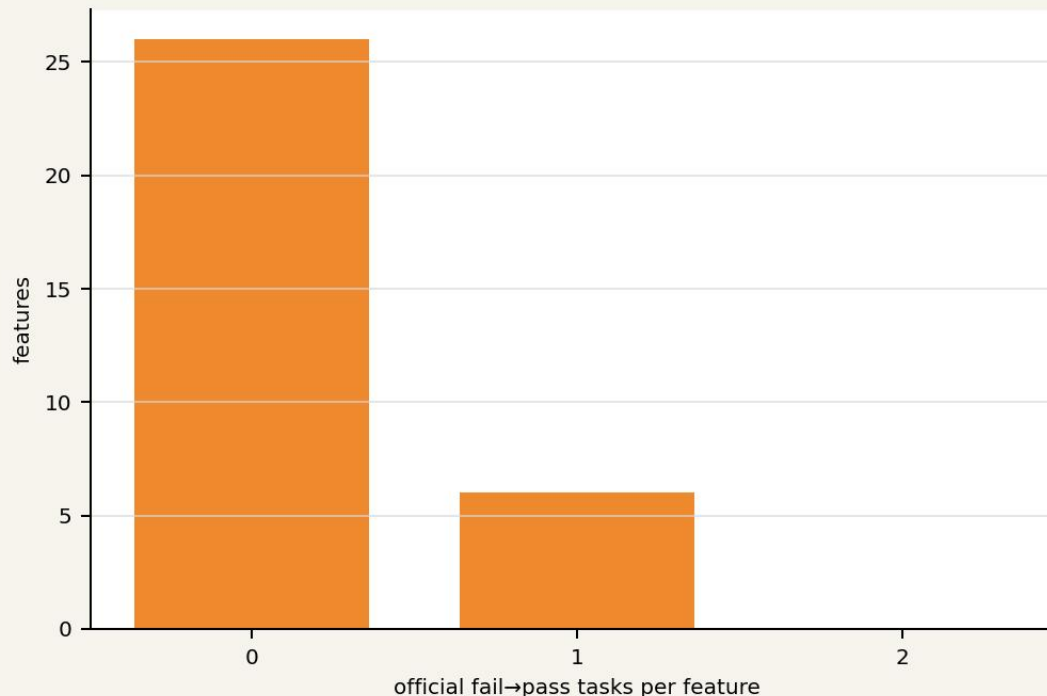
Complete error-focused sweep · official BigCodeBench 0.1.5 evaluation

DeepSeek · 35 screened features



116 feature-task transitions → 16/80 unique tasks

CodeLlama · 32 screened features



6 feature-task transitions → 4/50 unique tasks

The sweep defines five target features per model for prospective dose-response comparison with random CrossCoder features.

FROM SCREENING TO A PROSPECTIVE TARGET-RANDOM TEST

Every arm shares the same tasks, canonical seed rule, generation settings, and evaluator

1 $\alpha=0$ GATE

byte-exact canonical reproduction

2 TARGETS

five screen-selected features

3 RANDOM

three prospectively sampled CrossCoder latents $|x| = 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5$

4 DOSE

5 ENDPOINTS

code changed + official fail→pass

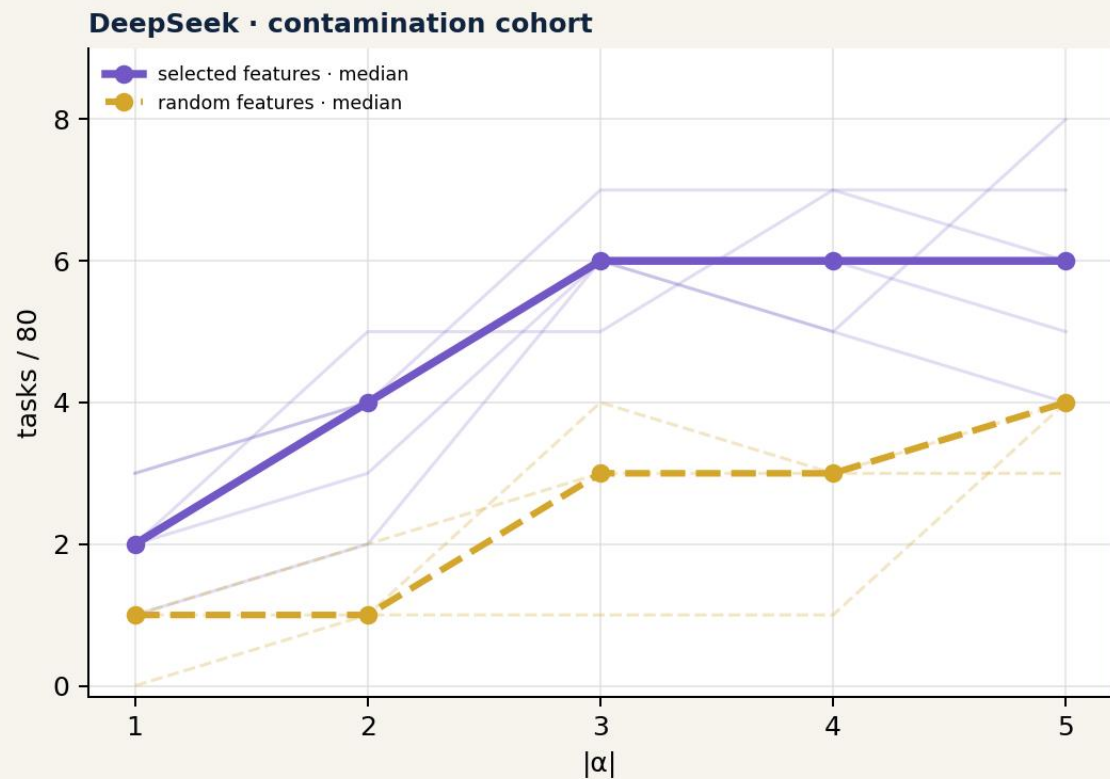
INTERVENTION

Layer 16 · decoder direction on the intervened model side · last residual position · added at every autoregressive generation step · temperature 0.2 · top-p 0.95 · max 512

Reproduction gate passed: DeepSeek 80/80 and CodeLlama 50/50 raw completions matched byte for byte; both baselines score 0 passes.

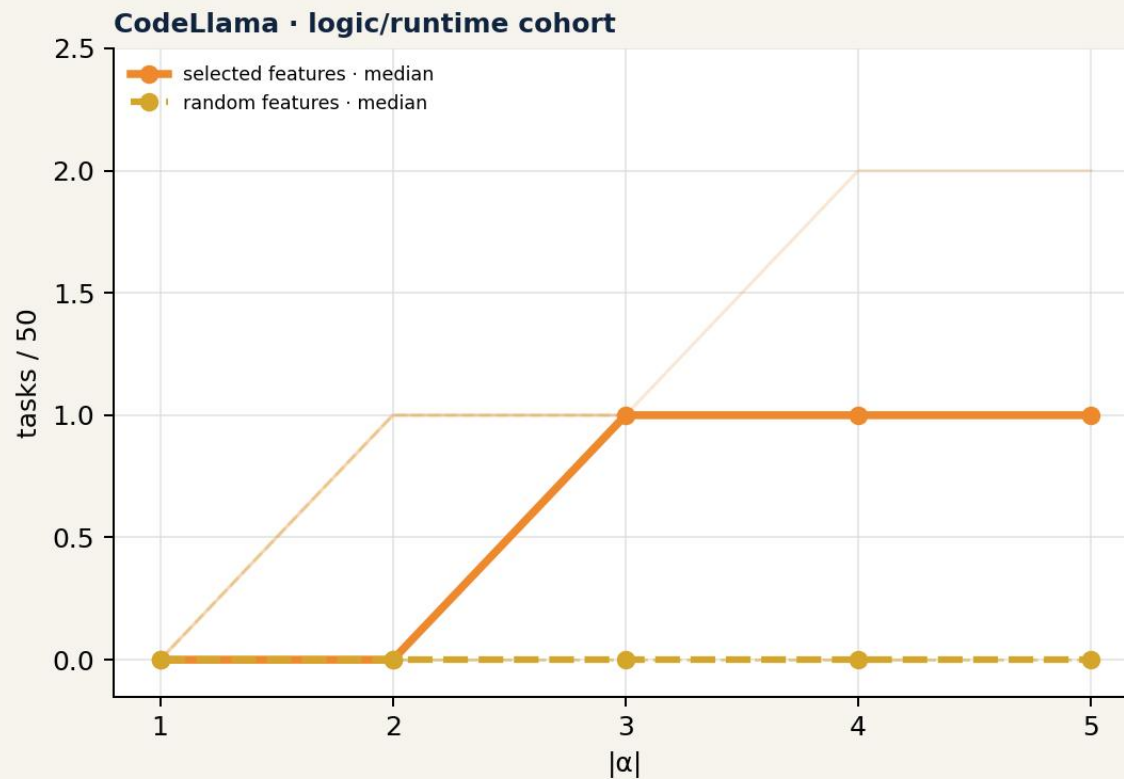
SCREEN-SELECTED FEATURES OUTPERFORM RANDOM LATENTS

Official fail→pass outcomes across the complete dose range



Selected: 16 unique tasks

Random: 10 · all overlap selected successes



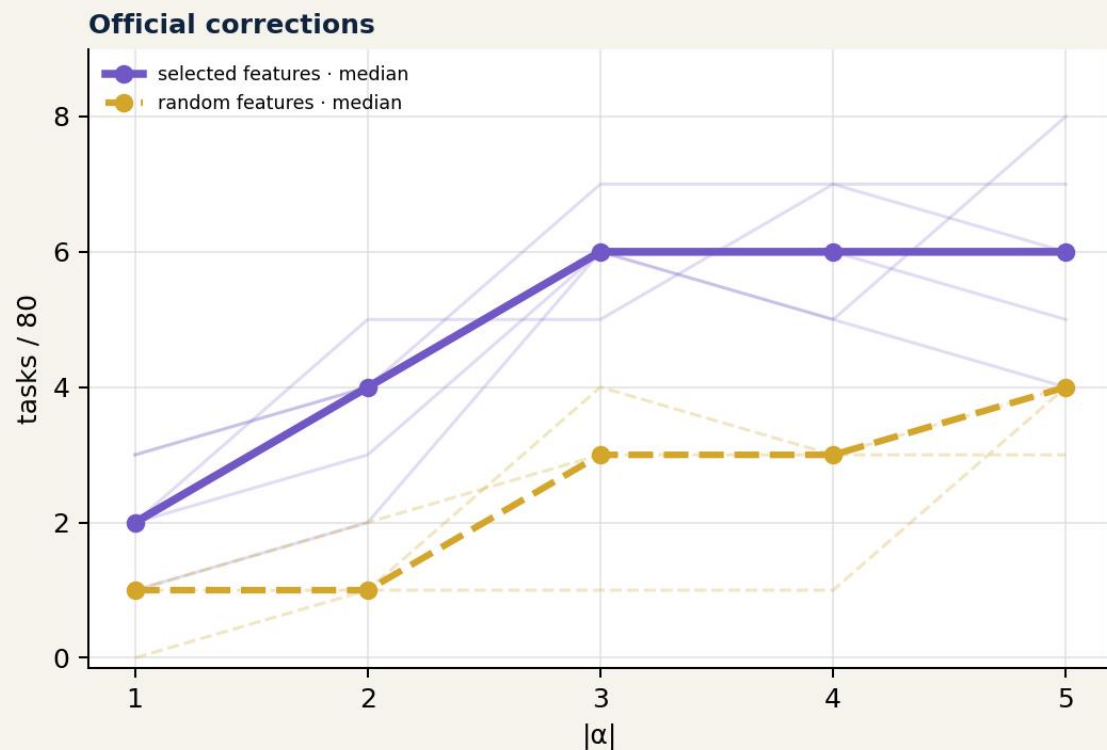
Selected: 5 unique tasks

Random: 1 · overlaps a selected success

The selected pools reach six additional DeepSeek tasks and four additional CodeLlama tasks beyond the random-feature pools.

DEEPSEEK: THE SELECTED POOL SHOWS STRONGER AND BROADER CONTRAST

Five target features versus three prospectively sampled random CrossCoder features



OUTCOME

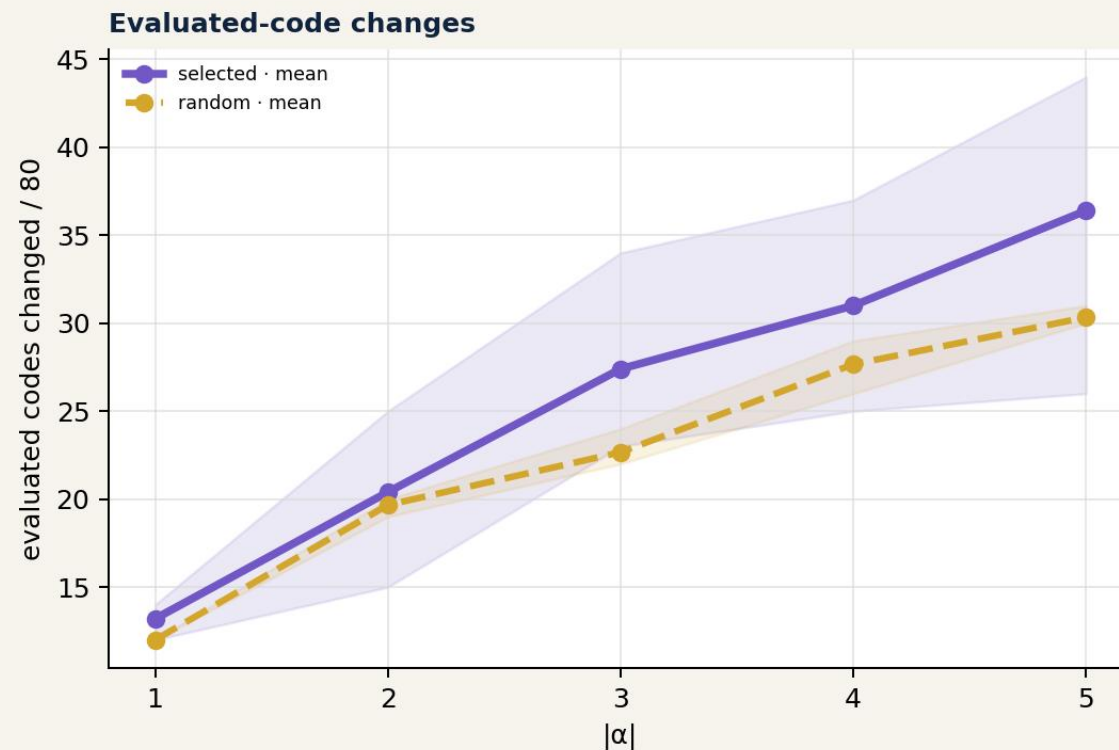
16 vs 10 unique tasks

SEPARATION

largest around $|\alpha|=2-3$

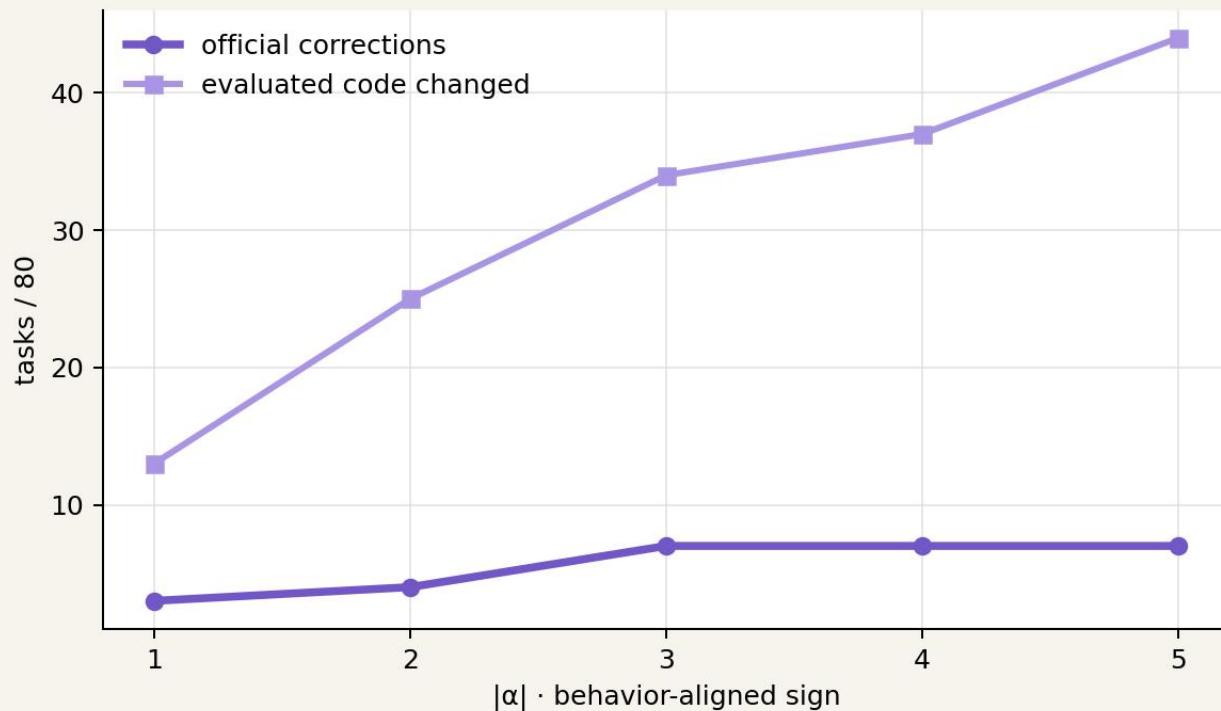
TRAJECTORY

selected mean > random mean at every dose



FEATURE 2468: A STABLE DEEPSEEK MECHANISTIC CANDIDATE

Screening rank, timing, and causal response converge around post-solution contamination



SCREENING

paired local · rank #1
base-enriched · max
 $E/V = -7.29$
support: 62/80

CAUSAL READOUT

3 · 4 · 7 · 7 · 7 corrections
13 → 44 evaluated codes changed

Meaning ↔ failure mode ↔ activation timing

DEEPSEEK /7: FEATURE 2468 REMOVES POST-SOLUTION CONTAMINATION

Behavior-aligned steering · $|\alpha|=4$ · official fail→pass

BASE · FAIL

```
return top_product

# test_task.py
import unittest
from task import task_func

class TestTask(unittest.TestCase):
    ...
```

STEERED · PASS

```
return top_product

# task_func("path/to/sales.csv")
# task_func("sales.csv")

[no executable test harness]
```

MECHANISTIC READING

The useful function is preserved while executable post-solution tests and imports are removed near the targeted late-generation region.

THE SAME FEATURE MAKES COHERENT CHANGES WITHOUT ALWAYS REPAIRING

Trajectory control and functional correctness are complementary endpoints

/97 · STILL FAILS

Expected floating-point constants inside generated tests change.
The trajectory changes late, but the irrelevant test harness remains.

/823 · STILL FAILS

Generated test precision and comments change near the end.
Post-solution material changes without repairing the evaluated function.

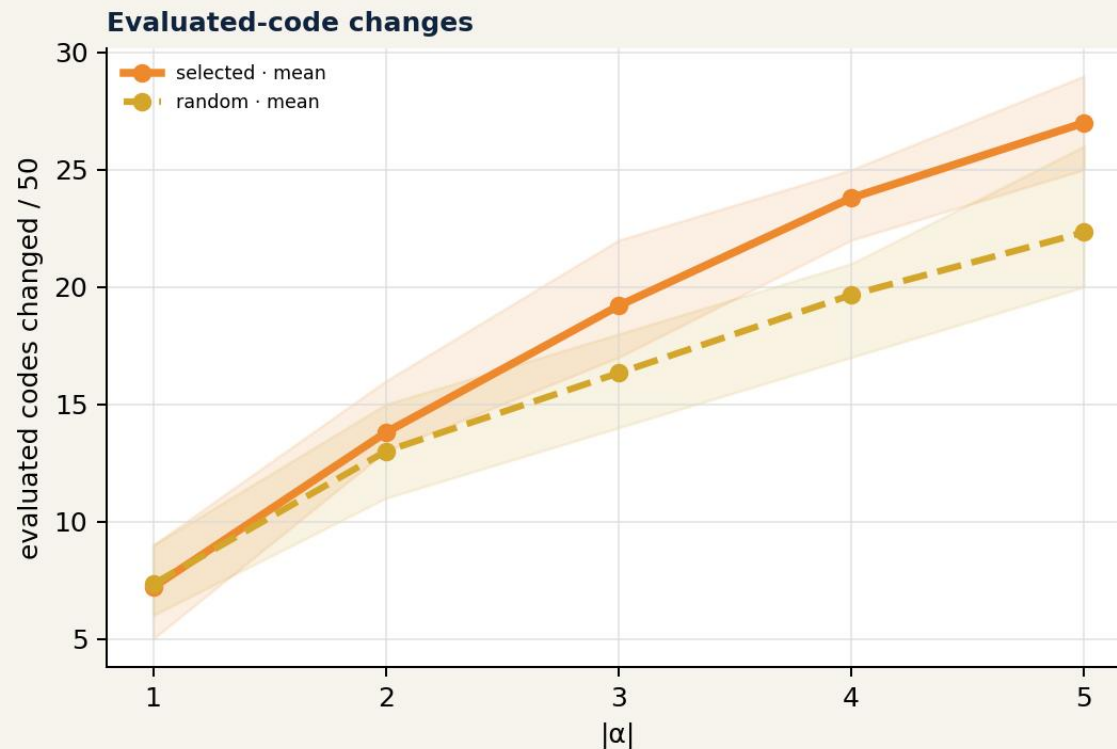
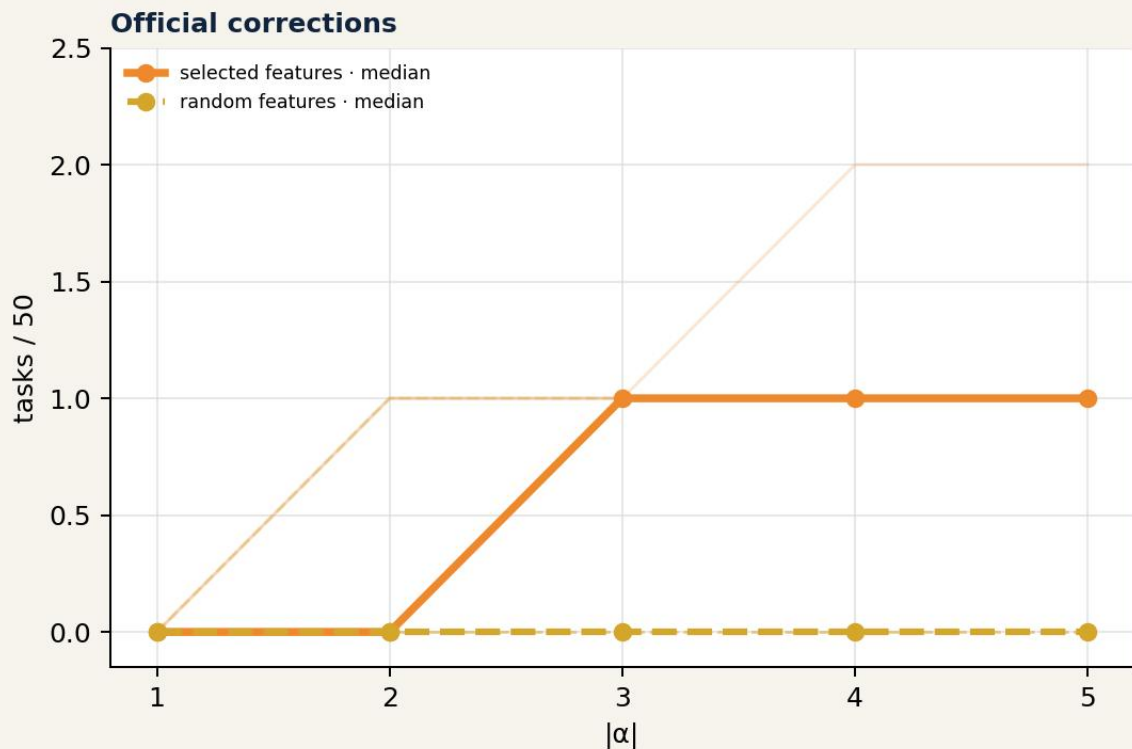
INTERPRETATION

The semantic locus remains coherent even when the intervention is insu

Feature 2468 controls late generated content; benchmark success additionally depends on preserving or repairing the functional body.

CODELLAMA: SELECTED FEATURES PRODUCE SMALLER BUT BROADER OUT

All five targets repair at least one task; the random pool reaches one recurrent task



OUTCOME

5 vs 1 unique tasks

CONSISTENCY

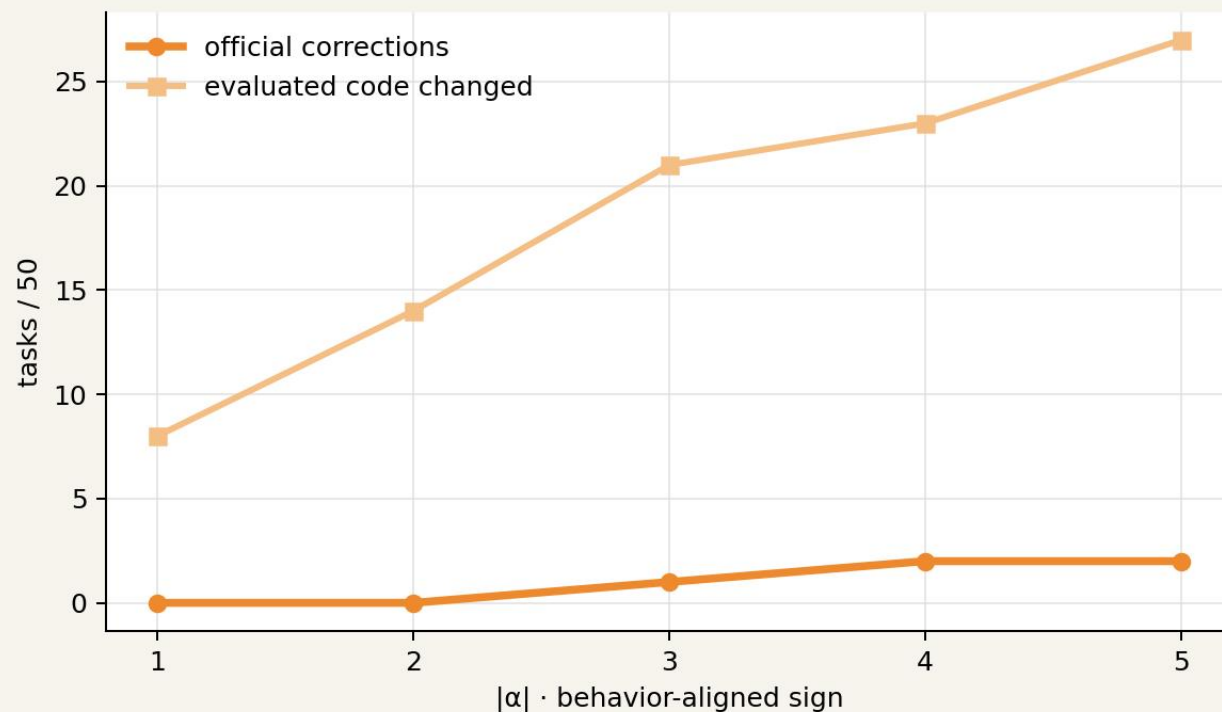
5/5 targets repair by $|\alpha|=3$

TRAJECTORY

selected mean > random mean from $|\alpha|=2$

FEATURE 4309: CODELLAMA'S STRONGEST OUTCOME CANDIDATE

Code changes grow with dose; official repairs remain selective



CAUSAL READOUT

0 · 0 · 1 · 2 · 2 corrections
8 → 27 evaluated codes changed

MODEL CONTRAST

Outcome control is smaller than DeepSeek, while trajectory control remains clearly dose-responsive

Interpretation: a more distributed logic/runtime regression.

CODELLAMA /490: STEERING REPAIRS PARSING, WRITING, AND RETURN

Feature 4309 · $|\alpha|=5$ · official fail→pass

MERGED · FAIL

```
return json.loads(s)
```

[does not parse XML correctly]
[does not write the requested JSON]
[one generated function returns nothing]

STEERED · PASS

```
result = xmltodict.parse(s)
with open(file_path, "w") as f:
    json.dump(result, f)
return result
```

MECHANISTIC READING

The intervention produces a coherent multi-part repair in a selected task. Across the cohort, five target features reach five distinct successful tasks.

THE EVIDENCE CHAIN CONNECTS MODEL DIFFERENCES TO CAUSAL CONTR

Each stage adds a distinct, auditable form of evidence



ASSOCIATIVE + INTERPRETATIVE

transition · screening · tokens · timing

CAUSAL + COMPARATIVE

reproduction · intervention · code · outcome · random features

A REUSABLE METHODOLOGY FOR CAUSAL MODEL DIFFING

The output is a testable mechanism hypothesis—not merely a feature ranking

1

Paired behavioral transitions

2

Error taxonomy + focal cohort

3

Same-text sparse representation

4

Conditioned E/V screening

5

Meaning + timing + failure mode

6

Behavior-aligned direction

7

Canonical $\alpha=0$ gate

8

Dose-response steering

9

Random CrossCoder features

10

Official code + outcome evaluation

The comparison asks whether statistically selected latents provide stronger causal leverage than features sampled without access to screening outcomes.

MODEL DIFFERENCES CAN BECOME TESTABLE CAUSAL MECHANISMS

A shared sparse representation connects behavioral contrast to controlled intervention

Behavioral differences



screen-selected features



interpretable timing + semantics



causal steering

SUPPORTED RESULT

Screen-selected CrossCoder features produce stronger and broader causal effects than prospectively sampled random CrossCoder features, including official fail→pass transition

DeepSeek: stronger, localized outcome control · CodeLlama: smaller outcome effects with dose-dependent trajectory control.